ARDUINO

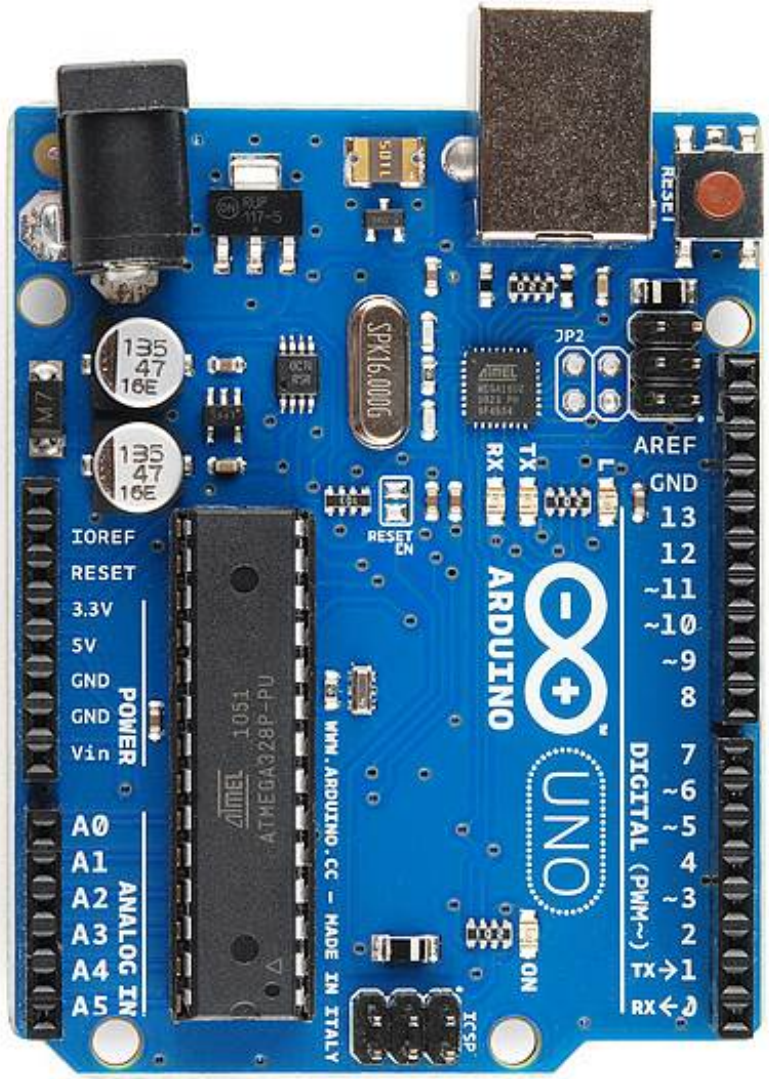
Alan L. Vergara

I CAN

- create simple prototypes on sensor and measurement technology.

Review:

- Open Source electronic prototyping platform based on flexible easy to use hardware and software.



ARDUINO FAMILY



Arduino Uno



Arduino Leonardo



Arduino Mega ADK



Arduino Ethernet



Arduino Due



Arduino Yún



Arduino Mega 2560

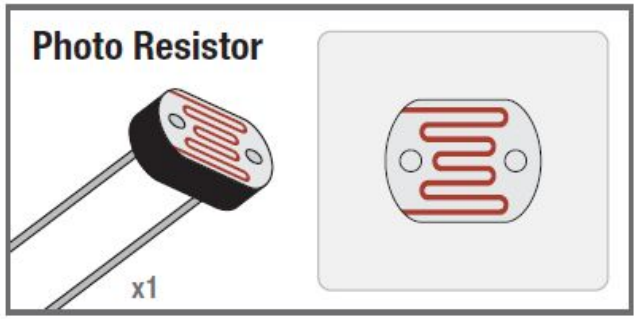


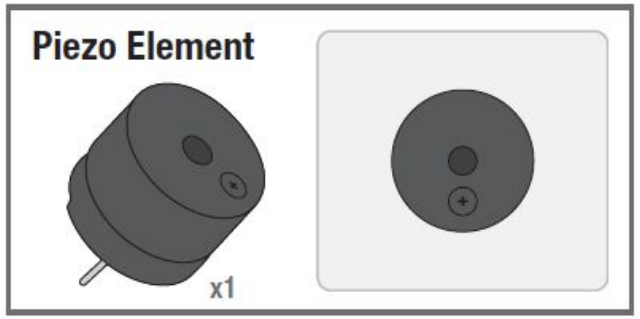
Arduino Mini

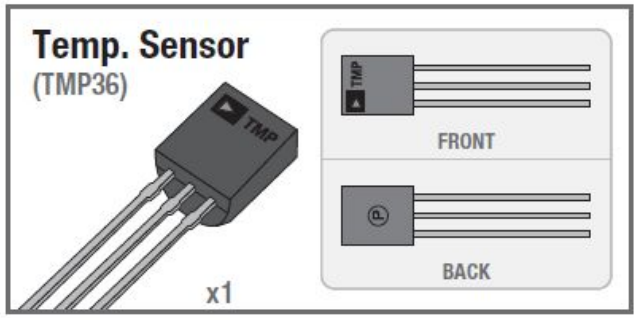
ARDUINO SIK COMPONENTS

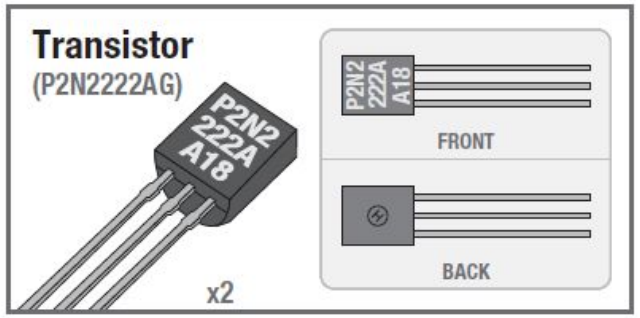
Name	Image	Type	Function	Notes
Push Button		Digital Input	Switch - Closes or opens circuit	Polarized, needs resistor
Trim potentiometer		Analog Input	Variable resistor	Also called a Trimpot.
Photoresistor		Analog Input	Light Dependent Resistor (LDR)	Resistance varies with light.
Relay		Digital Output	Switch driven by a small signal	Used to control larger voltages
Temp Sensor		Analog Input	Temp Dependent Resistor	
Flex Sensor		Analog Input	Variable resistor	
Soft Trimpot		Analog Input	Variable resistor	Careful of shorts
RGB LED		Dig & Analog Output	16,777,216 different colors	Ooh... So pretty.

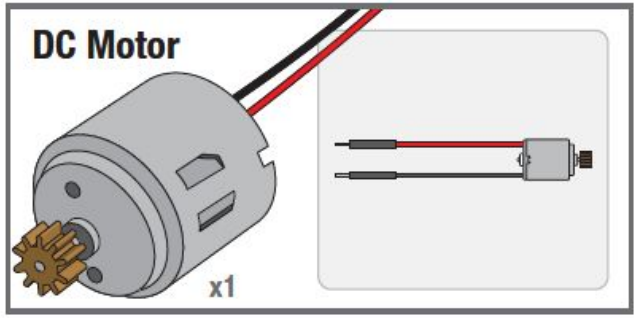
ARDUINO SIK COMPONENTS

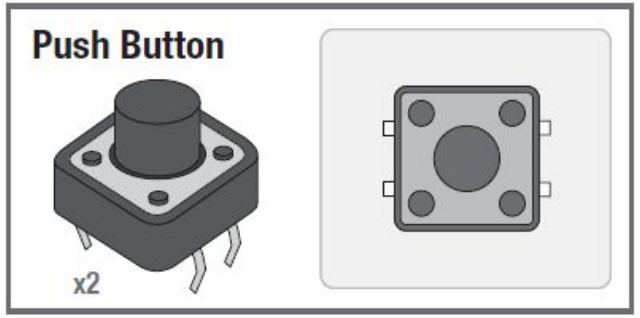


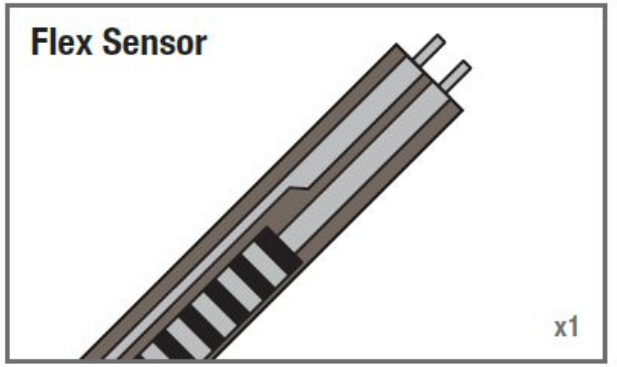


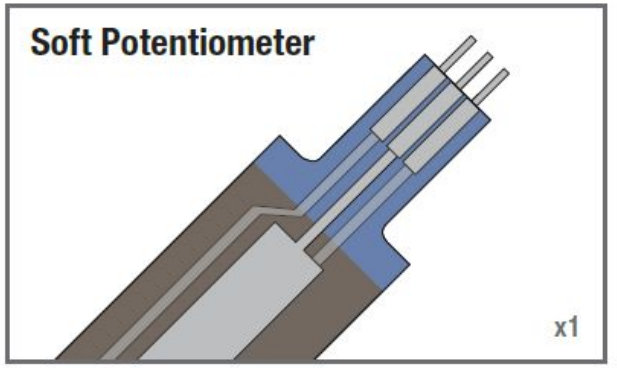


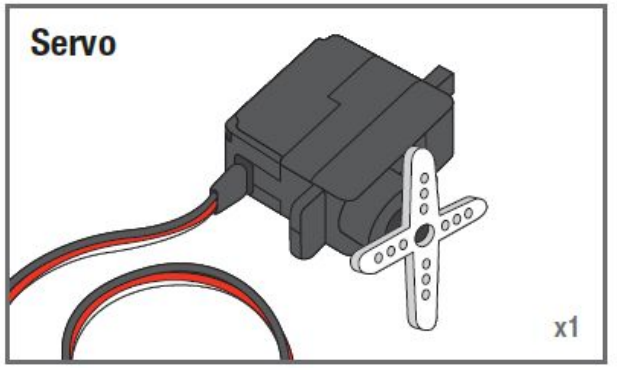












Bare minimum code

```
void setup() {
// put your setup code here, to run once:
}

void loop(){
// put your main code here, to run repeatedly:
}
```

Bare minimum code

- setup : It is called only when the Arduino is powered on or reset. It is used to initialize variables and pin modes

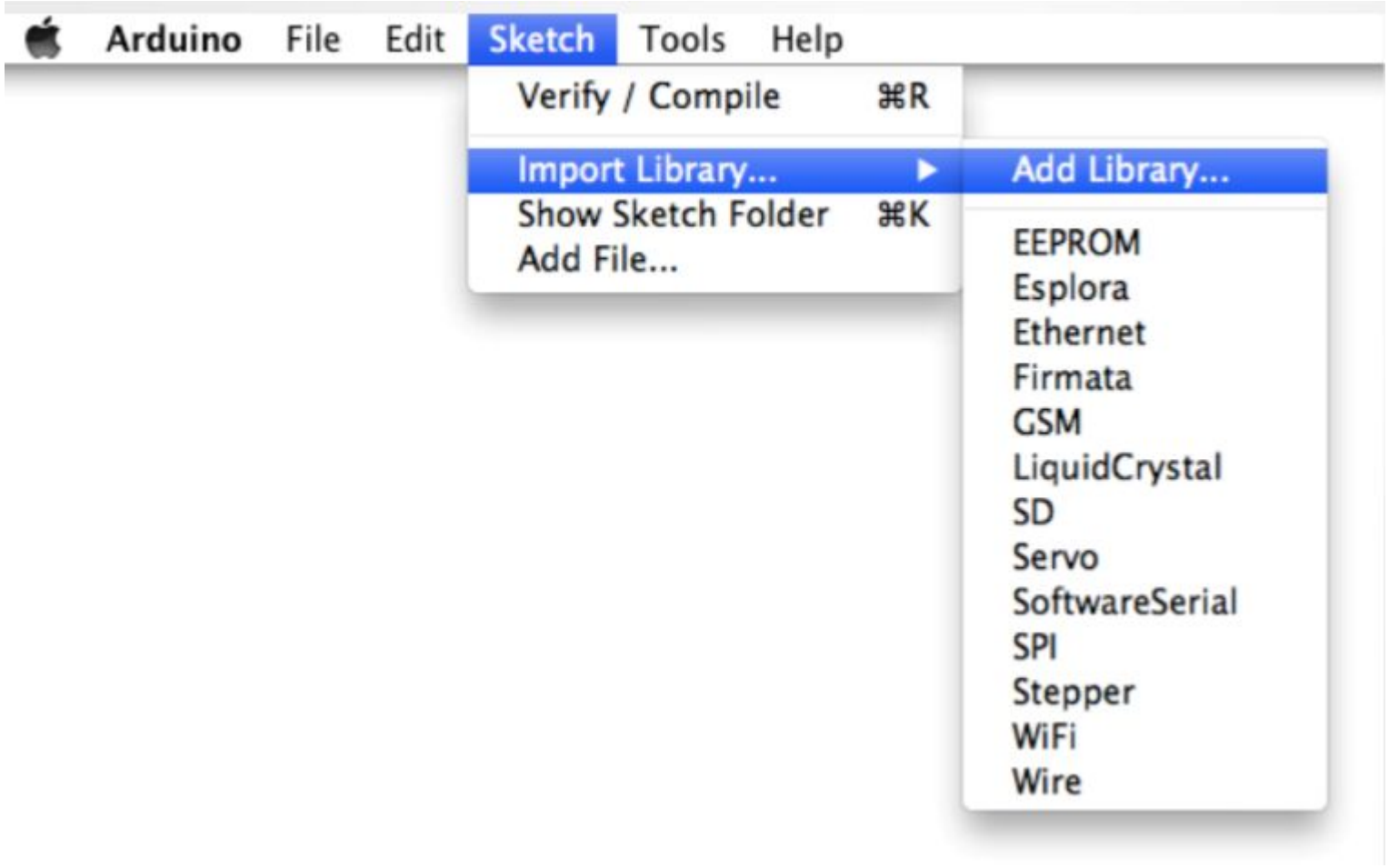
- loop : The loop functions runs continuously till the device is powered off. The main logic of the code goes here. Similar to while (1) for micro-controller programming.

LIBRARIES

LIBRARIES

- Libraries are a collection of code that makes it easy for you to connect to a sensor, display, module, etc. For example, the built-in LiquidCrystal library makes it easy to talk to character LCD displays. There are hundreds of additional libraries available on the Internet for download.

How to use LIBRARIES?



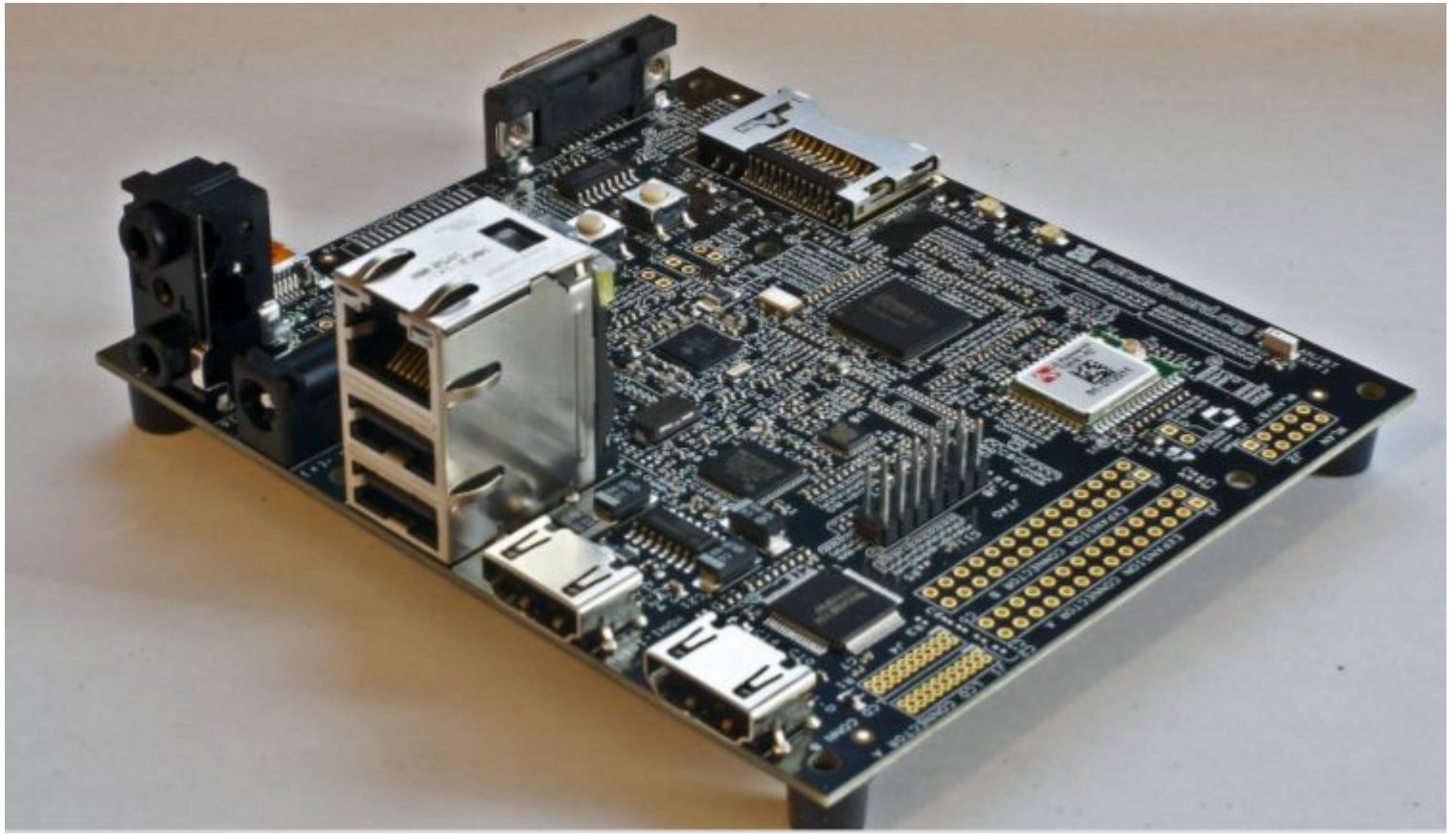
Single Board Devices

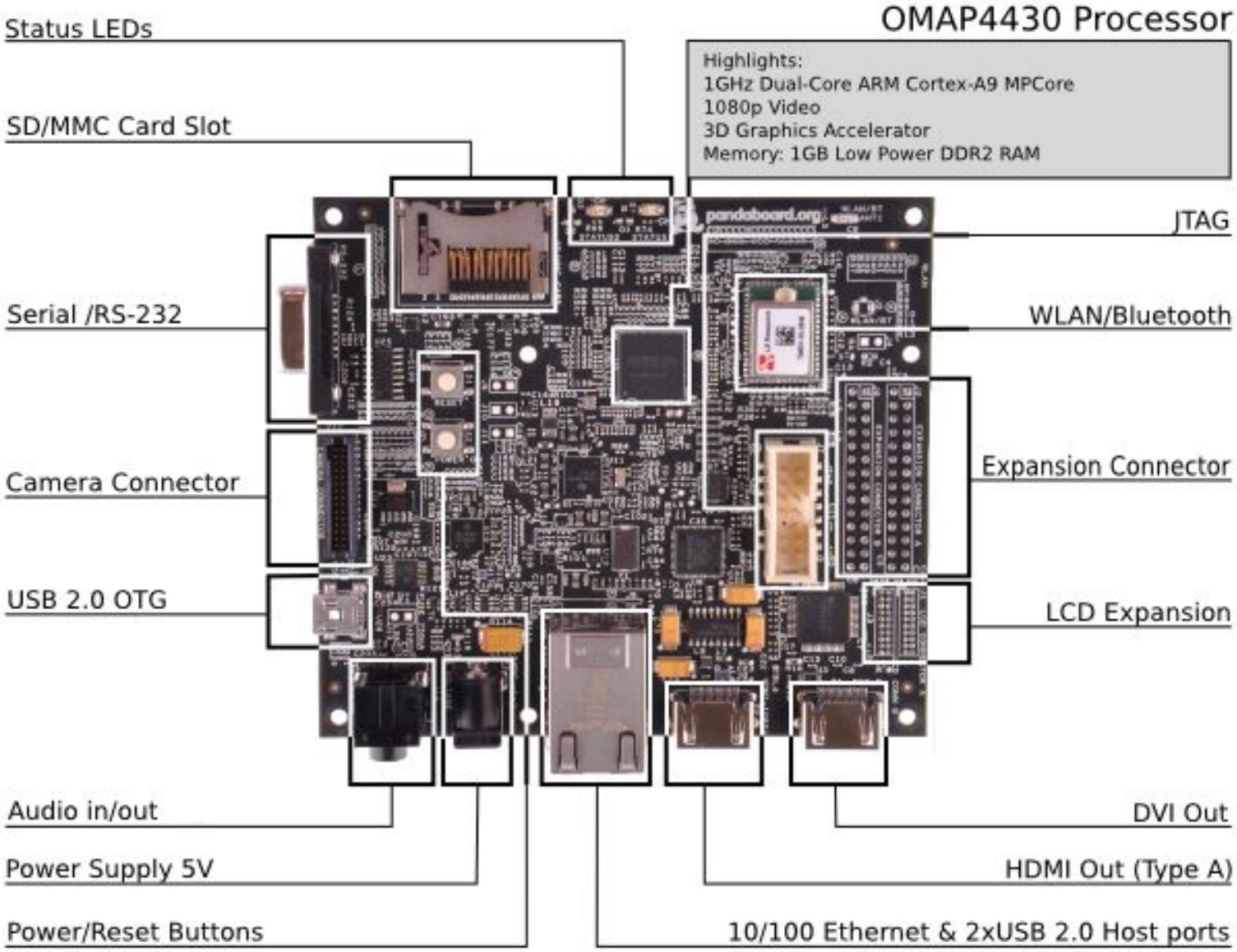
- Early microcomputers typically consisted of a half dozen (or more) circuit boards--plugged into a backplane--that implemented the central processor unit (CPU), memory, disk controllers and serial/parallel port functions. These backplane-based microcomputers were used for data acquisition, process control and R&D projects, but were generally too bulky to be used as embedded systems within devices.



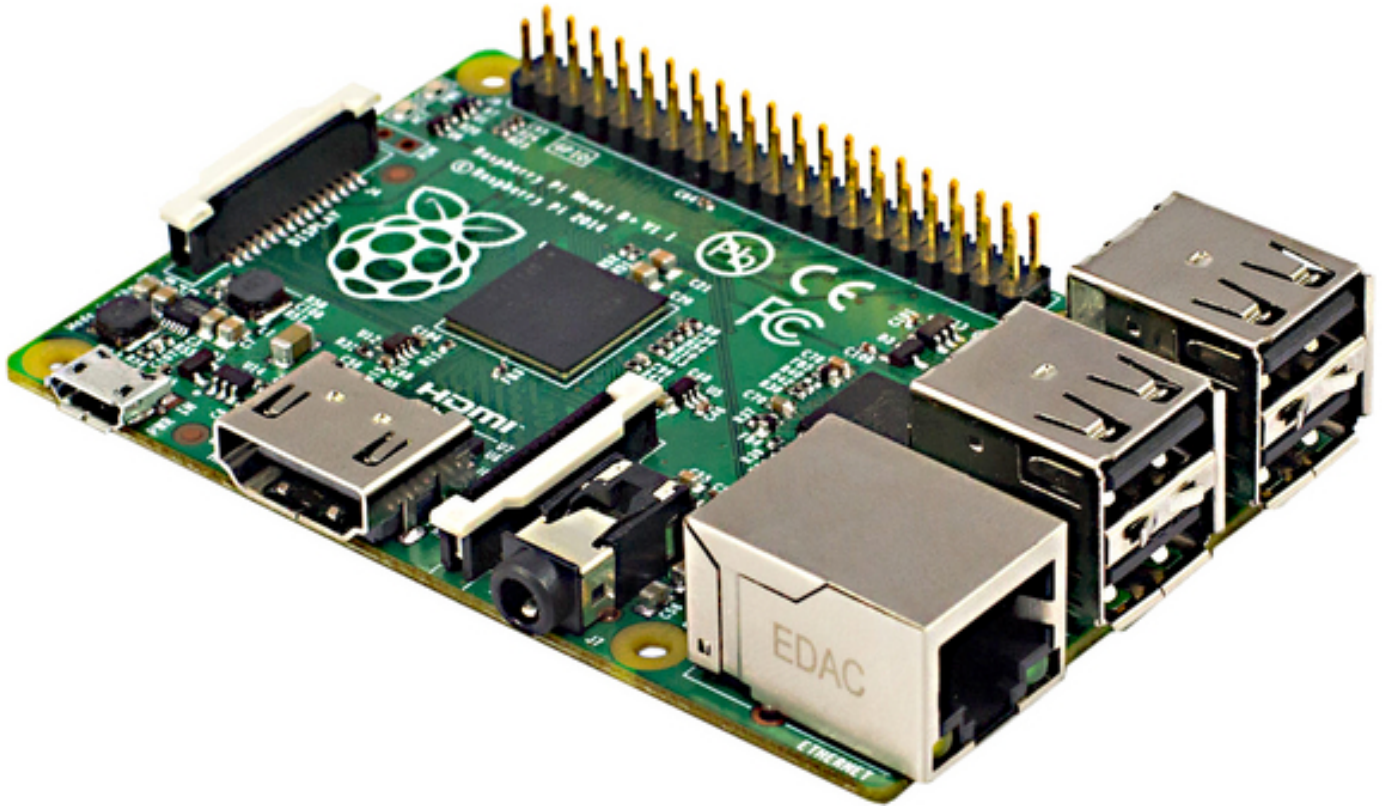
Example f Single Board Devices

- Pandaboard

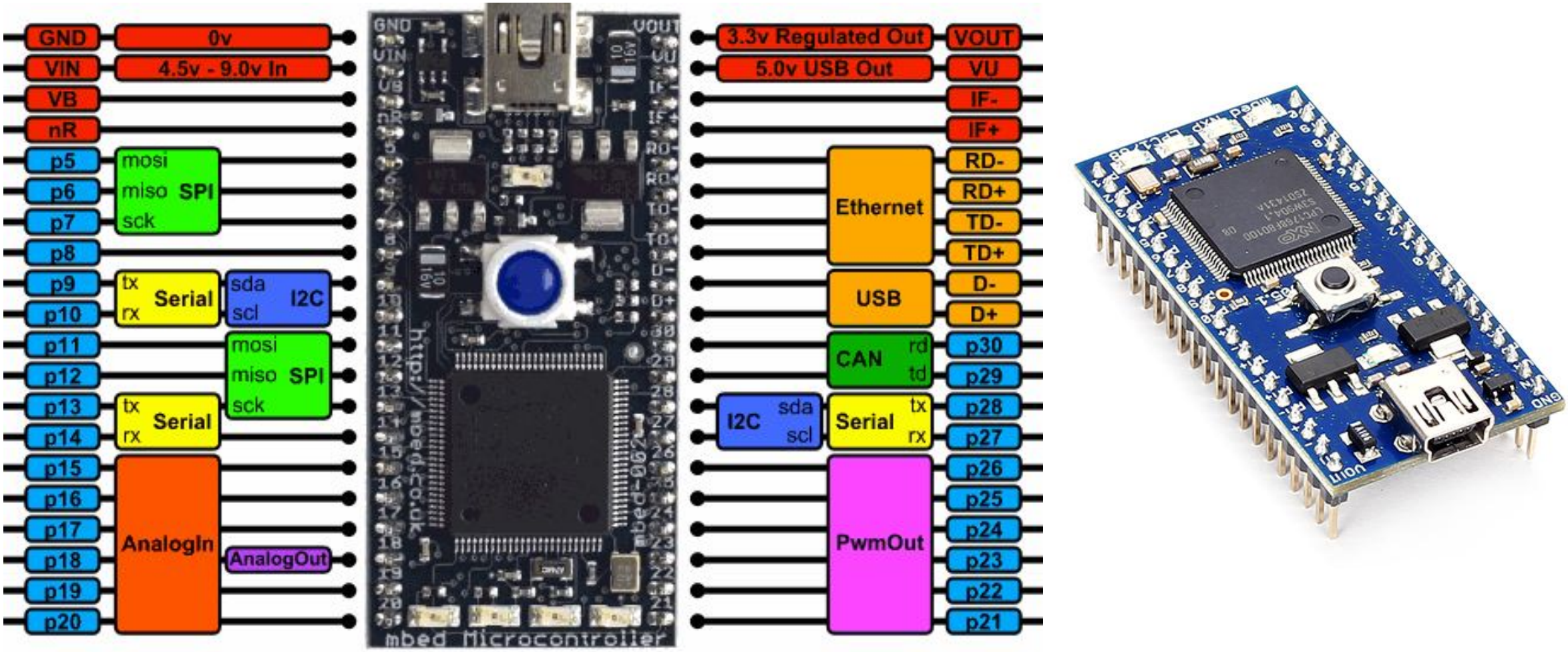



Status LEDs
SD/MMC Card Slot
Serial /RS-232
Camera Connector
USB 2.0 OTG
Audio in/out
Power Supply 5V
Power/Reset Buttons
OMAP4430 Processor
Highlights:
1GHz Dual-Core ARM Cortex-A9 MPCore
1080p Video
3D Graphics Accelerator
Memory: 1GB Low Power DDR2 RAM
JTAG
WLAN/Bluetooth
Expansion Connector
LCD Expansion
DVI Out
HDMI Out (Type A)
10/100 Ethernet & 2xUSB 2.0 Host ports
Board Dimensions: W:4.0" (101.6 mm) X H: 4.5" (114.3 mm)

RASPBERRY PI

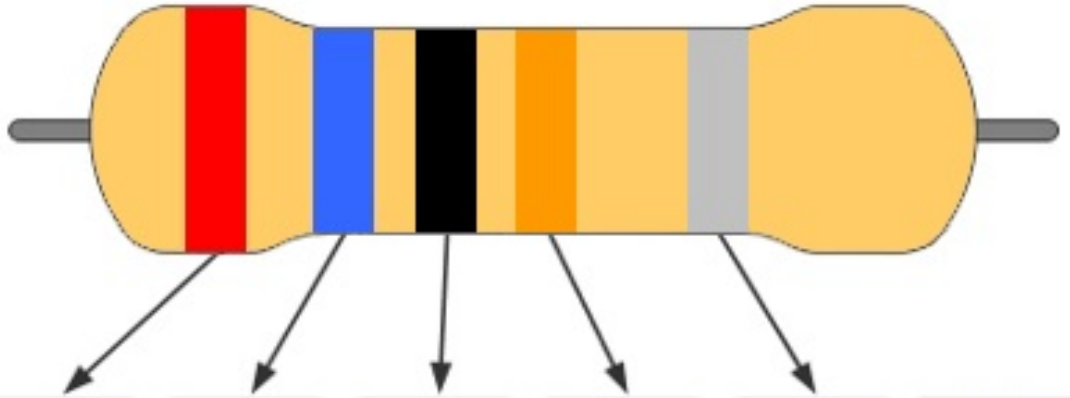


Mbed


GND
0v
VIN
4.5v - 9.0v In
VB
nR
p5
mosi
miso
SPI
sck
p6
p7
p8
p9
tx
rx
Serial
sda
scl
I2C
p10
p11
mosi
miso
SPI
sck
p12
p13
tx
rx
Serial
p14
p15
p16
p17
AnalogIn
AnalogOut
p18
p19
p20
http://mbed.co.uk
mbed Microcontroller
3.3v Regulated Out
VOUT
5.0v USB Out
VU
IF-
IF+
RD-
RD+
TD-
TD+
Ethernet
USB
D-
D+
CAN
rd
td
p30
p29
I2C
sda
scl
Serial
tx
rx
p28
p27
PwmOut
p26
p25
p24
p23
p22
p21

RESISTOR COLOR CODES

RESISTOR COLOR CODES



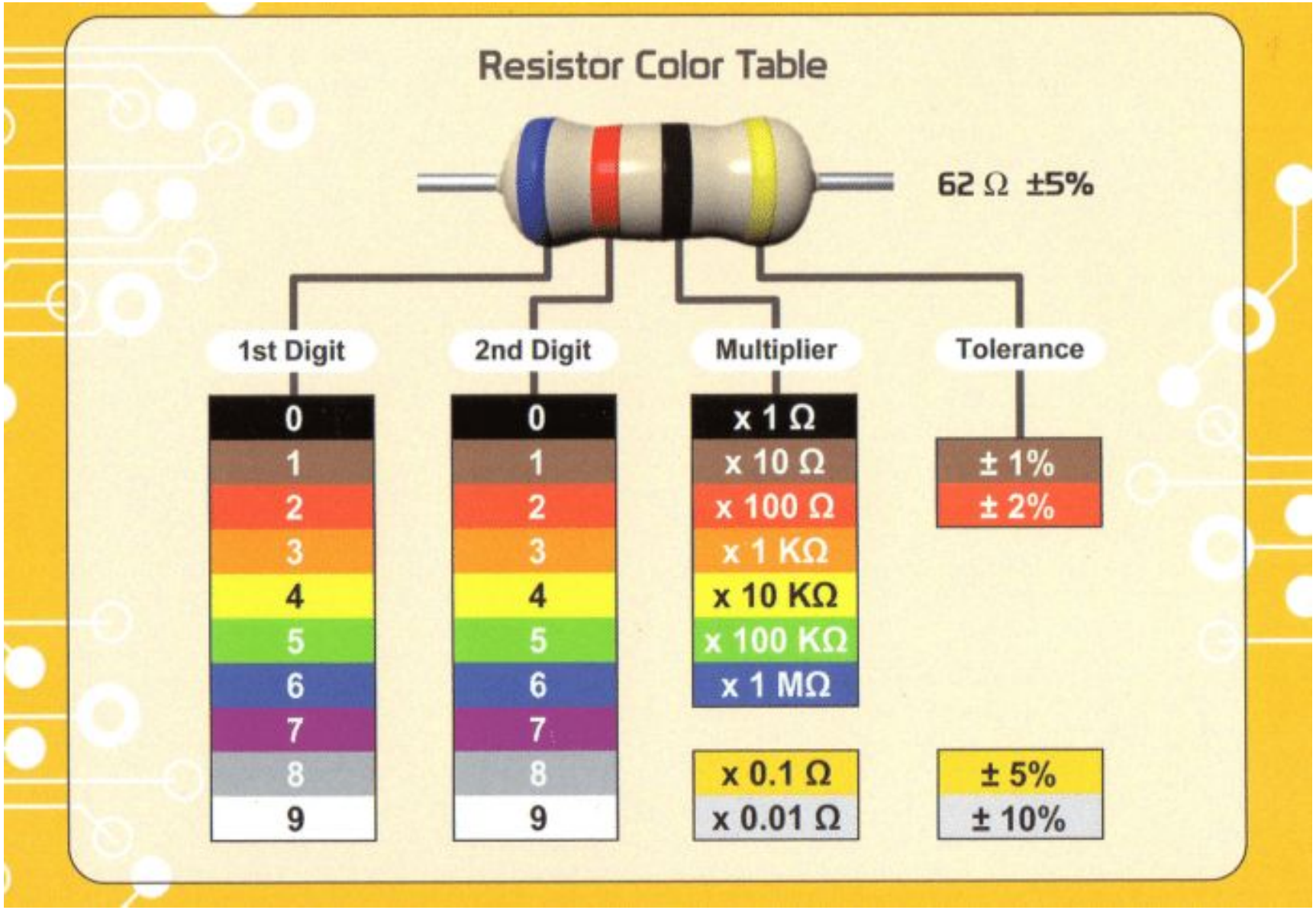
	1st digit	2nd digit	3rd digit	multiply	tolerance	TCR (ppm/K)
Black	0	0	0	1	1% (F)	100
Brown	1	1	1	10	2% (G)	50
Red	2	2	2	100		15
Orange	3	3	3	1K		25
Yellow	4	4	4	10K		
Green	5	5	5	100K	0.5% (D)	
Blue	6	6	6	1M	0.25% (C)	10
Violet	7	7	7	10M	0.1% (B)	5
Gray	8	8	8	100M	0.05% (A)	
White	9	9	9	1G		
Gold				0.1	5% (J)	
Silver				0.01	10% (K)	
None					20% (M)	

Resistor Color Codes

- What is the values for your resistors?
- Use the Color Codes to figure it out
- Use also a Multimeter to see if you get the same results.



Example


Resistor Color Table
62 Ω ±5%
1st Digit
2nd Digit
Multiplier
Tolerance
0
1
2
3
4
5
6
7
8
9
x 1 Ω
x 10 Ω
x 100 Ω
x 1 KΩ
x 10 KΩ
x 100 KΩ
x 1 MΩ
x 0.1 Ω
x 0.01 Ω
± 1%
± 2%
± 5%
± 10%

SENSORS AND ACTUATORS





Push Button

Temperature Sensor

Light Sensor

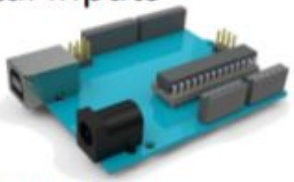




Potentiometer

Analog/Digital Inputs

Sensors



Signals from the surroundings

LED



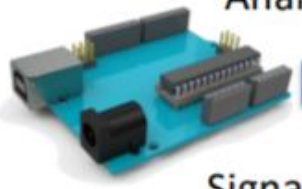


Motor

Buzzer



Analog/Digital Outputs



Actuators

Signals to the surroundings

Sensors and Actuators

- A **Sensor** is a converter that measures a physical size and converts it to a signal that can be read by an instrument, data acquisition device, or an Arduino. Examples: temperature sensor, pressure sensor, etc.

Sensors and Actuators

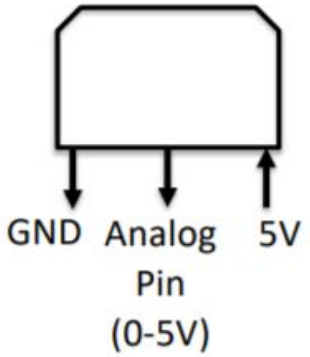
- An **Actuator** is a kind of motor that moves or controls a mechanism or system. It is powered by an energy source, typical electric current, hydraulic fluid pressure, or air pressure, and converts this energy into motion. Examples: Engine, Pump, Valve, etc.

Sensors and Actuators

- The sensors and actuators can be either digital or analog.
- Some sensors and actuators have been made for Arduino, while others need to be connected in some circuit to work properly with Arduino.
- Many of these come with ready-made libraries for Arduino, so they are easy to use.

Potentiometer

- A potentiometer is a simple knob that provides a variable resistance, which we can read into the Arduino board as an analog value.





INDIVIDUAL ASSESSMENT

- Create a simple traffic light using Arduino UNO on Tinkercad

- What you need:
- 1 Arduino Uno R3
- 1 Red LED
- 1 Yellow LED
- 1 Green LED
- 3 220 Ω Resistor

